Newman-Penrose Formalism Calculations

Stephani et al., Equation (33.12_2)

Metric

$$
\begin{aligned}
ds^2 = & \left(\frac{2\left(6\,a^2\epsilon x - \left(3\,A\epsilon m r + 2\left(Aa^2\epsilon r^2 - Aa^2\epsilon\right)k\left(r\right)\right)e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\left(Ba^2\epsilon r^2 - Ba^2\epsilon\right)k\left(r\right)\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2}\right) \mathrm{d}u \otimes \mathrm{d}u - \mathrm{d}u \otimes \mathrm{d}r - \mathrm{d}r \otimes \mathrm{d}u \\
& + \left(\frac{B^2\epsilon^2 r^4 - 2\,B^2\epsilon^2 r^2 + B^2\epsilon^2 + \left(A^2\epsilon^2 r^4 - 2\,A^2\epsilon^2 r^2 + A^2\epsilon^2\right)e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\left(AB\epsilon^2 r^4 - 2\,AB\epsilon^2 r^2 + AB\epsilon^2\right)e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{x^6}\right) \mathrm{d}x \otimes \mathrm{d}x \\
& + \left(\frac{B^2\epsilon^2 r^4 - 2\,B^2\epsilon^2 r^2 + B^2\epsilon^2 + \left(A^2\epsilon^2 r^4 - 2\,A^2\epsilon^2 r^2 + A^2\epsilon^2\right)e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\left(AB\epsilon^2 r^4 - 2\,AB\epsilon^2 r^2 + AB\epsilon^2\right)e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{x^6}\right) \mathrm{d}y \otimes \mathrm{d}y
\end{aligned}
$$

Coordinates

$(x^\mu) = (u, r, x, y)$

Null Tetrad

Covariant components

$l = (1)\ du$

$$
n = \left(2\,\epsilon r^2 k\left(r\right) + \frac{3\,A\epsilon m r e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{\left(Ae^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B\right)a^2} - 2\,\epsilon k\left(r\right) - \frac{6\,\epsilon x}{Ae^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B}\right)\ du + (1)\ dr
$$

$$
m = \left(\frac{\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{2\,x^3} + \frac{\sqrt{2}B\epsilon r^2}{2\,x^3} - \frac{\sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{2\,x^3} - \frac{\sqrt{2}B\epsilon}{2\,x^3}\right)\ dx + \left(\frac{i\,\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{2\,x^3} + \frac{i\,\sqrt{2}B\epsilon r^2}{2\,x^3} - \frac{i\,\sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{2\,x^3} - \frac{i\,\sqrt{2}B\epsilon}{2\,x^3}\right)\ dy
$$

$$
\overline{m} = \left(\frac{\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{2\,x^3} + \frac{\sqrt{2}B\epsilon r^2}{2\,x^3} - \frac{\sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{2\,x^3} - \frac{\sqrt{2}B\epsilon}{2\,x^3}\right)\ dx + \left(-\frac{i\,\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{2\,x^3} - \frac{i\,\sqrt{2}B\epsilon r^2}{2\,x^3} + \frac{i\,\sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{2\,x^3} + \frac{i\,\sqrt{2}B\epsilon}{2\,x^3}\right)\ dy
$$

Contravariant components

$l = (-1)\ dr$

$$n = (-1)\ du + \left(\frac{2\,Aa^2\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} + \frac{2\,Ba^2\epsilon r^2 k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} - \frac{2\,Aa^2\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} + \frac{3\,A\epsilon m r e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} - \frac{2\,Ba^2\epsilon k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} - \frac{6\,a^2\epsilon x}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2}\right)\ dr$$

$$m = \left(\frac{x^3}{\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B\epsilon r^2 - \sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - \sqrt{2}B\epsilon}\right)\ dx + \left(\frac{i\,x^3}{\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B\epsilon r^2 - \sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - \sqrt{2}B\epsilon}\right)\ dy$$

$$\overline{m} = \left(\frac{x^3}{\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B\epsilon r^2 - \sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - \sqrt{2}B\epsilon}\right)\ dx + \left(-\frac{i\,x^3}{\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B\epsilon r^2 - \sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - \sqrt{2}B\epsilon}\right)\ dy$$

Directional Derivatives

$D = -\partial_r$

$$\Delta = -\partial_u + \frac{2\,Aa^2\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} + \frac{2\,Ba^2\epsilon r^2 k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} - \frac{2\,Aa^2\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} + \frac{3\,A\epsilon m r e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} - \frac{2\,Ba^2\epsilon k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} - \frac{6\,a^2\epsilon x}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2}\,\partial_r$$

$$\delta = \frac{x^3}{\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B\epsilon r^2 - \sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - \sqrt{2}B\epsilon}\,\partial_x + \frac{i\,x^3}{\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B\epsilon r^2 - \sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - \sqrt{2}B\epsilon}\,\partial_y$$

$$\overline{\delta} = \frac{x^3}{\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B\epsilon r^2 - \sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - \sqrt{2}B\epsilon}\,\partial_x - \frac{i\,x^3}{\sqrt{2}A\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B\epsilon r^2 - \sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - \sqrt{2}B\epsilon}\,\partial_y$$

Spin Coefficients

$$\rho = \frac{2\,r}{r^2 - 1}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{4\,Aa^2\epsilon r^3 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} k\left(r\right)}{Aa^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2r^2 - Aa^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - Ba^2} + \frac{4\,Ba^2\epsilon r^3 k\left(r\right)}{Aa^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2r^2 - Aa^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - Ba^2} - \frac{4\,Aa^2\epsilon r e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} k\left(r\right)}{Aa^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2r^2 - Aa^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - Ba^2}$$
$$+ \frac{3\,A\epsilon m r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{Aa^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2r^2 - Aa^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - Ba^2} - \frac{4\,Ba^2\epsilon r k\left(r\right)}{Aa^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2r^2 - Aa^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - Ba^2}$$
$$- \frac{12\,a^2\epsilon r x}{Aa^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2r^2 - Aa^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - Ba^2} + \frac{3\,A\epsilon m e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{Aa^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2r^2 - Aa^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - Ba^2}$$

$$\lambda = 0$$

$$\nu = -\frac{6\,x^3}{\sqrt{2}A^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,\sqrt{2}ABr^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B^2r^2 - \sqrt{2}A^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,\sqrt{2}ABe^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - \sqrt{2}B^2}$$

$$\alpha = \frac{3\,x^2}{2\left(\sqrt{2}A\epsilon r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B\epsilon r^2 - \sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - \sqrt{2}B\epsilon\right)}$$

$$\beta = -\frac{3\,x^2}{2\left(\sqrt{2}A\epsilon r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B\epsilon r^2 - \sqrt{2}A\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - \sqrt{2}B\epsilon\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$\epsilon = 0$

$$\gamma = -\frac{Aa^2\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}\frac{\partial}{\partial r}k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} - \frac{2\,Aa^2\epsilon r e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} - \frac{Ba^2\epsilon r^2\frac{\partial}{\partial r}k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} - \frac{2\,Ba^2\epsilon r k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} + \frac{Aa^2\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}\frac{\partial}{\partial r}k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} + \frac{Ba^2\epsilon\frac{\partial}{\partial r}k\left(r\right)}{Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2} - \frac{3\,A\epsilon m e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}}{2\left(Aa^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + Ba^2\right)}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{2}{r^2 - 1}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned}
\Phi_{11} = {} & \frac{A^2a^2\epsilon^3r^6e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}\frac{\partial^2}{(\partial r)^2}k\left(r\right)}{2\left(A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2r^4 - 2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2a^2\epsilon^2r^2 + A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2\right)} \\
& + \frac{ABa^2\epsilon^3r^6e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}\frac{\partial^2}{(\partial r)^2}k\left(r\right)}{A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2r^4 - 2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2a^2\epsilon^2r^2 + A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2} \\
& + \frac{2\,A^2a^2\epsilon^3r^5e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}\frac{\partial}{\partial r}k\left(r\right)}{A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2r^4 - 2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2a^2\epsilon^2r^2 + A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2} \\
& + \frac{4\,ABa^2\epsilon^3r^5e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}\frac{\partial}{\partial r}k\left(r\right)}{A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2r^4 - 2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2a^2\epsilon^2r^2 + A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2} \\
& + \frac{B^2a^2\epsilon^3r^6\frac{\partial^2}{(\partial r)^2}k\left(r\right)}{2\left(A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2r^4 - 2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2a^2\epsilon^2r^2 + A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2\right)} \\
& - \frac{3\,A^2a^2\epsilon^3r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}k\left(r\right)}{A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2r^4 - 2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2a^2\epsilon^2r^2 + A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^2\epsilon^2} - \cdots
\end{aligned}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\begin{aligned}
\Phi_{22} = &- \frac{8\,A^2a^4\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}k\left(r\right)^2}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} \\
&- \frac{16\,ABa^4\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}k\left(r\right)^2}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} \\
&- \frac{8\,B^2a^4\epsilon^2r^4k\left(r\right)^2}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} \\
&+ \frac{16\,A^2a^4\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}k\left(r\right)^2}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} \\
&+ \frac{32\,ABa^4\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}k\left(r\right)^2}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} \\
&- \frac{6\,A^2a^2\epsilon^2mr^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}\frac{\partial}{\partial r}k\left(r\right)}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} \\
&- \frac{6\,ABa^2\epsilon^2mr^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}\frac{\partial}{\partial r}k\left(r\right)}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} \\
&- \frac{12\,A^2a^2\epsilon^2mr^3e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}k\left(r\right)}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} \\
&- \frac{12\,ABa^2\epsilon^2mr^3e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}k\left(r\right)}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} \\
&+ \frac{48\,Aa^4\epsilon^2r^2xe^{\left(\frac{3\,\epsilon m u}{a^2}\right)}k\left(r\right)}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} \\
&+ \frac{16\,B^2a^4\epsilon^2r^2k\left(r\right)^2}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} \\
&+ \frac{48\,Ba^4\epsilon^2r^2xk\left(r\right)}{A^2a^4r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^4r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2a^4r^2 - A^2a^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 2\,ABa^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - B^2a^4} - \ldots
\end{aligned}$$

$$\Lambda = -\frac{A^2a^2\epsilon^3r^6e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}\frac{\partial^2}{(\partial r)^2}k\left(r\right)}{6\left(A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2r^4-2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}-4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}-2\,B^2a^2\epsilon^2r^2+A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2\right)}$$

$$-\frac{ABa^2\epsilon^3r^6e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}\frac{\partial^2}{(\partial r)^2}k\left(r\right)}{3\left(A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2r^4-2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}-4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}-2\,B^2a^2\epsilon^2r^2+A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2\right)}$$

$$-\frac{2\,A^2a^2\epsilon^3r^5e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}\frac{\partial}{\partial r}k\left(r\right)}{A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2r^4-2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}-4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}-2\,B^2a^2\epsilon^2r^2+A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2}$$

$$-\frac{4\,ABa^2\epsilon^3r^5e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}\frac{\partial}{\partial r}k\left(r\right)}{A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2r^4-2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}-4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}-2\,B^2a^2\epsilon^2r^2+A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2}$$

$$-\frac{B^2a^2\epsilon^3r^6\frac{\partial^2}{(\partial r)^2}k\left(r\right)}{6\left(A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2r^4-2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}-4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}-2\,B^2a^2\epsilon^2r^2+A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2\right)}$$

$$-\frac{17\,A^2a^2\epsilon^3r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}k\left(r\right)}{3\left(A^2a^2\epsilon^2r^4e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2r^4e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2r^4-2\,A^2a^2\epsilon^2r^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}-4\,ABa^2\epsilon^2r^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}-2\,B^2a^2\epsilon^2r^2+A^2a^2\epsilon^2e^{\left(\frac{6\,\epsilon m u}{a^2}\right)}+2\,ABa^2\epsilon^2e^{\left(\frac{3\,\epsilon m u}{a^2}\right)}+B^2a^2\epsilon^2\right)}-\cdots$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$\Psi_2 = \frac{A^2 a^2 \epsilon^3 r^6 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} \frac{\partial^2}{(\partial r)^2} k\left(r\right)}{3\left(A^2 a^2 \epsilon^2 r^4 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 r^4 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2 r^4 - 2\,A^2 a^2 \epsilon^2 r^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2 r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2 a^2 \epsilon^2 r^2 + A^2 a^2 \epsilon^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2\right)}$$

$$+ \frac{2\,ABa^2 \epsilon^3 r^6 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} \frac{\partial^2}{(\partial r)^2} k\left(r\right)}{3\left(A^2 a^2 \epsilon^2 r^4 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 r^4 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2 r^4 - 2\,A^2 a^2 \epsilon^2 r^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2 r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2 a^2 \epsilon^2 r^2 + A^2 a^2 \epsilon^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2\right)}$$

$$+ \frac{B^2 a^2 \epsilon^3 r^6 \frac{\partial^2}{(\partial r)^2} k\left(r\right)}{3\left(A^2 a^2 \epsilon^2 r^4 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 r^4 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2 r^4 - 2\,A^2 a^2 \epsilon^2 r^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2 r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2 a^2 \epsilon^2 r^2 + A^2 a^2 \epsilon^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2\right)}$$

$$- \frac{2\,A^2 a^2 \epsilon^3 r^4 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} k\left(r\right)}{3\left(A^2 a^2 \epsilon^2 r^4 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 r^4 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2 r^4 - 2\,A^2 a^2 \epsilon^2 r^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2 r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2 a^2 \epsilon^2 r^2 + A^2 a^2 \epsilon^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2\right)}$$

$$- \frac{4\,ABa^2 \epsilon^3 r^4 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} k\left(r\right)}{3\left(A^2 a^2 \epsilon^2 r^4 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 r^4 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2 r^4 - 2\,A^2 a^2 \epsilon^2 r^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2 r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2 a^2 \epsilon^2 r^2 + A^2 a^2 \epsilon^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2\right)}$$

$$- \frac{A^2 a^2 \epsilon^3 r^4 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} \frac{\partial^2}{(\partial r)^2} k\left(r\right)}{A^2 a^2 \epsilon^2 r^4 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 r^4 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2 r^4 - 2\,A^2 a^2 \epsilon^2 r^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,ABa^2\epsilon^2 r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^2 a^2 \epsilon^2 r^2 + A^2 a^2 \epsilon^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,ABa^2\epsilon^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^2 a^2 \epsilon^2} - \dots$$

$$\Psi_3 = -\frac{12\,rx^3}{\sqrt{2}A^2 r^4 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,\sqrt{2}ABr^4 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B^2 r^4 - 2\,\sqrt{2}A^2 r^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 4\,\sqrt{2}ABr^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,\sqrt{2}B^2 r^2 + \sqrt{2}A^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 2\,\sqrt{2}ABe^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + \sqrt{2}B^2}$$

$$\Psi_4 = -\frac{18\,x^5}{A^3\epsilon r^4 e^{\left(\frac{9\,\epsilon m u}{a^2}\right)} + 3\,A^2 B\epsilon r^4 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 3\,AB^2\epsilon r^4 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^3\epsilon r^4 - 2\,A^3\epsilon r^2 e^{\left(\frac{9\,\epsilon m u}{a^2}\right)} - 6\,A^2 B\epsilon r^2 e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} - 6\,AB^2\epsilon r^2 e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} - 2\,B^3\epsilon r^2 + A^3\epsilon e^{\left(\frac{9\,\epsilon m u}{a^2}\right)} + 3\,A^2 B\epsilon e^{\left(\frac{6\,\epsilon m u}{a^2}\right)} + 3\,AB^2\epsilon e^{\left(\frac{3\,\epsilon m u}{a^2}\right)} + B^3\epsilon}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"